Structural count bounds beyond the small-noise comparison

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The direct comparison through $ns = .075$ is retained unchanged. This note supplies two tools for the remaining noise range: a concentration bound from the ordered conditional counts, and an analytic upper bound for the signed binomial curvature correction. The first gives a complete comparison in the symmetric central subcase with no noise ceiling. The tools do not yet close the general larger-noise comparison.

1 Concentration from the ordered counts

Let $Y = B + U$, where $\mathbb{P}(B = 1) = p \in (0, 1)$, $\mathbb{E}(U \mid B) = 0$, and

$$U_0 \le a_0, \qquad U_1 \ge -a_1, \qquad a_0, a_1 \ge 0, \quad \zeta = 1 - a_0 - a_1 > 0.$$

Take independent copies U_{0i}, U_{1i} and set

$$T_j = j + \sum_{i \le j} U_{1i} + \sum_{i > j} U_{0i}, \qquad \pi_j = \binom{n}{j} p^j (1-p)^{n-j}, \quad 0 \le j \le n.$$

Every conditional count has the correct law, and $T_{j+1} - T_j \ge \zeta$ on each coupled sample.

Theorem 1. *For a closed interval I of length less than $m\zeta$, where $1 \le m \le n+1$ is an integer,*

$$\mathbb{P}(Y_1 + \cdots + Y_n \in I) \le \max_{0 \le j \le n+1-m} \sum_{r=j}^{j+m-1} \pi_r.$$

In particular every atom, and every interval of length less than ζ, has probability at most $\max_j \pi_j$.

Proof. For each coupled sample, the indices with $T_j \in I$ are consecutive, and at most m of them can occur: $m + 1$ points would have span at least $m\zeta$. Conditioning on the actual binomial count gives

$$\mathbb{P}(Y_1 + \cdots + Y_n \in I) = \mathbb{E} \sum_{j=0}^{n} \pi_j \mathbf{1}_{\{T_j \in I\}}.$$

A shorter block can be extended to m legal indices, so its mass is bounded by the displayed maximum. This proves the claim without any independence assertion about a selected index. The strict length condition matters: for noiseless Bernoulli variables, $[j, j+1]$ contains two atoms. □

Theorem 2 (Symmetric central comparison at every noise level)**.** *Let n be even, $p = 1/2$, U_1 have the law of $-U_0$, $\mathbb{E}U_0 = 0$, $U_0 \le a < 1/2$, and assume finite third absolute moments. Set $s = \mathbb{E}U_0^2 > 0$, $X = (Y - 1/2)/\sqrt{1/4 + s}$ and $\rho = \mathbb{E}|X|^3$. For every $C > 0$ and*

$$0 \le H < \frac{1 - 2a}{2\sqrt{1/4 + s}},$$

the direct comparison with the central Bernoulli anchor $k = n/2$ is strict:

$$\sqrt{n}\{\mathbb{P}(X_1 + \cdots + X_n \le H) - \Phi(H/\sqrt{n})\} - C\rho \\ - \sqrt{n}\{F_B(n/2) - 1/2\} + C < 0.$$

There is no upper bound on ns in this statement.

Proof. The marginal X is symmetric. For $H \ge 0$ this gives the exact identity

$$\mathbb{P}(X_1 + \cdots + X_n \le H) = \frac{1}{2} + \frac{1}{2}\mathbb{P}(|X_1 + \cdots + X_n| \le H).$$

The corresponding raw interval for the sum of the Y_i has length $2H\sqrt{1/4 + s} < 1 - 2a$. Theorem 1 bounds its mass by the central binomial mass $\pi_{n/2}$. Also $F_B(n/2) = 1/2 + \pi_{n/2}/2$. Subtraction therefore bounds the displayed comparison by

$$-\sqrt{n}\{\Phi(H/\sqrt{n}) - 1/2\} - C(\rho - 1).$$

Jensen's strict inequality gives $\rho > 1$ unless $|X| = 1$ almost surely. The two conditional supports lie strictly on opposite sides of zero, since $a < 1/2$. Thus $|X| = 1$ would force each conditional law to be a point mass and $s = 0$, contrary to the hypothesis. The final bound is strict, including at $H = 0$ and at exact atom collisions. □

2 The signed curvature correction has a short uniform bound

Write $v_j = \mathbb{E}U_j^2$, $q = 1 - p$, $v = pq$, $s = qv_0 + pv_1$, and $W_j = T_j - j$. For any real t, define

$$B_j(t) = \mathbb{E}(t - T_j)_+ - (t - j)_+ \ge 0, \qquad \mathcal{C}(t) = \sum_{j=1}^{n-1} (\pi_{j+1} - \pi_j) B_j(t).$$

These quantities require only finite second moments. The new crossing identity in the companion `crossing.tex` writes the actual CDF as the linear interpolation of the binomial CDF, plus $\mathcal{C}(t)$, minus a weighted nonnegative crossing loss. The following estimate bounds only the correction; it is not a bound for that loss.

Lemma 3. *For $n \ge 2$, $0 < p < 1$ and every real t, put $c = (np - q - t)_+$. Then*

$$\mathcal{C}(t) \le \frac{s}{4v} + \frac{c\sqrt{s}}{2v\sqrt{n+1}}. \tag{1}$$

In terms of $\lambda = ns$, this is

$$\sqrt{n}\,\mathcal{C}(t) \le \frac{\lambda}{4v\sqrt{n}} + \frac{c\sqrt{\lambda}}{2v\sqrt{n+1}}.$$

The second term vanishes whenever $t \ge np - q$.

Proof. Let $a = t - j$ and $t_j = (n - j)v_0 + jv_1$. Centering gives

$$B_j(t) = \frac{\mathbb{E}|a - W_j| - |a|}{2}.$$

The triangle inequality and Cauchy–Schwarz imply $B_j \le \sqrt{t_j}/2$. When $a > 0$, another application of Cauchy–Schwarz gives

$$B_j \le \frac{\sqrt{a^2 + t_j} - a}{2} \le \frac{t_j}{4a}.$$

Thus $a_+ B_j \le t_j/4$ for all a. Write $\widetilde{\pi}_r = \mathbb{P}(\mathrm{Bin}(n+1,p) = r)$. The exact adjacent-mass identity is

$$\pi_{j+1} - \pi_j = \frac{(n+1)p - j - 1}{(n+1)v} \widetilde{\pi}_{j+1}.$$

Its numerator is $a + (np - q - t) \le a_+ + c$. Discard the nonpositive mass differences, using $B_j \ge 0$. On the positive side,

$$\mathcal{C}(t) \le \frac{1}{(n+1)v} \sum_j \widetilde{\pi}_{j+1} \left(\frac{t_j}{4} + \frac{c}{2}\sqrt{t_j} \right).$$

The remaining sums may be extended over all $1 \le j \le n-1$. For $r = j+1$, $t_j = (n+1-r)v_0 + (r-1)v_1$ is at most $(n+1-r)v_0 + rv_1$. Extending this latter nonnegative expression over the complete binomial distribution gives

$$\sum_{j=1}^{n-1} \widetilde{\pi}_{j+1} t_j \le (n+1)s.$$

Cauchy–Schwarz also bounds the corresponding square-root sum by $\sqrt{(n+1)s}$, since its total mass is at most one. Substitution proves (1). No local truncation of the binomial distribution, or numerical count grid, was used. □

The signed sum can be much smaller than this upper bound. The proof retains its favorable sign long enough to discard only the decreasing side of the mass function. Nevertheless, below $np - q$ the remaining square-root term is not a bound proportional to arbitrarily small noise. Even above $np - q$, a quantitative weighted crossing-loss estimate and the normal/cubic repair are still needed to prove the full comparison.

3 A precise obstruction to one four-moment shortcut

Under the companion crossing theorem's hypotheses, put $g_n(t) = t - \mathbb{E}K$. Its kernel satisfies $h(x) \ge h_D(x) := \mathbb{P}(D > x) - \mathbb{E}(D - x)_+$: below the lower endpoint of D, the right side is $x \le \mathbb{E}(x - U_0)_+$, and above it the two kernels agree. Selecting one index in the nonnegative sum therefore gives

$$g_n(t) \ge \mathbb{P}(W_i > t - i) - \mathbb{E}(W_i - (t-i))_+$$

by selecting one index. It would be useful if the expectation on the right were positive from the usual own-count moment bounds alone. This is false. Let

$$\mathbb{P}(Z = 6/5) = \mathbb{P}(Z = -6/5) = 25/144, \qquad \mathbb{P}(Z = 0) = 47/72.$$

Then

$$\mathbb{E}Z = \mathbb{E}Z^3 = 0, \quad \mathbb{E}Z^2 = 1/2, \quad \mathbb{E}Z^4 = 18/25 \le (3/5)^2(1/2) + 3(1/2)^2 = 93/100,$$

but

$$\mathbb{P}(Z > 0) - \mathbb{E}Z_+ = -5/144 < 0.$$

Thus no polynomial argument using only these four moment constraints can prove positivity of that selected reserve on the whole moderate-noise range. This law is a counterexample to the moment relaxation, not an asserted convolution of admissible iid conditional variables. Additional independence information, joint neighboring terms, or the full nonnegative kernel is necessary.

A second false shortcut concerns renewal independence. Take $A = 3/5$, $a = 3/125$ and the comonotone conditional coupling

$$(U_0, U_1) = (-A, -a), (a, -a), (a, A) \quad \text{with probabilities } 1/26, 12/13, 1/26.$$

Both marginals are centered and obey the right strip with inward sum $2a = .048$. At $n = 42, p = 1/2$ their noise is $ns = 42Aa = 378/625 = .6048$. Although the increment $D = 1 + U_1 - U_0$ has mean one and width $78/125 < 1$,

$$\mathrm{Cov}(U_0, D) = -216/15625 \neq 0.$$

The initial mesh location $\sum U_{0i}$ therefore remains correlated with its increments. An ordinary independent-delay renewal theorem cannot be applied merely by choosing this coupling. The companion exact stop-loss telescoping identity accounts for the bias and avoids the issue.

4 Evidence and the remaining inequality

The exact checker uses actual centered conditional two-point marginals, including one vanishing conditional noise and exact collisions. It verifies interval concentration, the binomial difference identity, the weighted variance step, the curvature bound and the symmetric central comparison. Radical bounds in the curvature control are checked by squaring only after checking the sign of the residual. The two counterexamples above are exact.

A separate bounded diagnostic tested 115 actual four/three-point laws at $n = 42, p = 1/2$ and noise $.075 < ns \leq .616$. It evaluated the unweighted crossing at 43,487 exact atom locations near the mean, with floating probabilities; all observed losses were positive. Those observations are discovery only and are not inputs to any theorem here.

The unresolved general implication is quantitative. In the exact identity

$$\Delta_C = -\sqrt{n} \sum_i \pi_i g_i(t) + \sqrt{n}\, \mathcal{C}(t) + \text{phase, normal and cubic repairs},$$

one must bound the first, weighted loss strongly enough for the same actual law to absorb every remaining term. The fact that $\sum_i g_i(t) \geq 0$, even for the whole acquired noise range, does not supply such a bound. The existing accepted count certificates remain in place; no larger-noise full theorem or global propagation is claimed.